

Microsoft Introduction to AI and Machine Learning

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HOMEWORK #4: MNIST

The MNIST database (Modified National Institute of Standards and Technology database) is a cornerstone dataset in machine learning and computer vision, often referred to as the "Hello World" of deep learning. It consists of 70,000 grayscale images of handwritten digits (0-9) that are widely used for training and testing image classification models.

MNIST Classification:

1. Do an SVD analysis of the digit images. You will need to reshape each image into a column vector and each column of your data matrix is a different image.
2. What does the singular value spectrum look like and how many modes are necessary for good image reconstruction? (i.e. what is the rank r of the digit space?)
3. What is the interpretation of the $\mathbf{U}$, $\mathbf{\Sigma}$, and $\mathbf{V}$ matrices?
4. On a 3D plot, project onto three selected $\mathbf{V}$ -modes (columns) colored by their digit label. For example, columns 2,3, and 5.

Extra: Once you have performed the above and have your data projected into PCA space, you will build a classifier to identify individual digits in the training set.

- Pick two digits. See if you can build a linear classifier (LDA) that can reasonably identify/classify them.
- Pick three digits. Try to build a linear classifier to identify these three now.
- Which two digits in the data set appear to be the most difficult to separate? Quantify the accuracy of the separation with LDA on the test data.
- Which two digits in the data set are most easy to separate? Quantify the accuracy of the separation with LDA on the test data.
- SVM (support vector machines) and decision tree classifiers were the state-of-the-art until about 2014. How well do these separate between all ten digits? (see code below to get started).
- Compare the performance between LDA, SVM and decision trees on the hardest and easiest pair of digits to separate (from above).

Make sure to discuss the performance of your classifier on both the training and test sets.

NOTE: You will write a narrative report about this homework on your github page.